

COSC 39: Theory of Computation

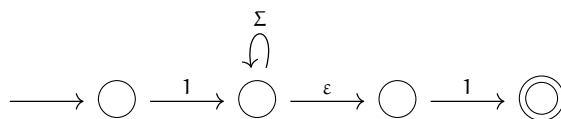
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Credit Statement: Talked to Carson Bates'27, and Sophie Park'27.

Problem 1. Construct an ϵ DFA to recognize the language

$$L_1 := \{w \in \{0, 1\}^* : w \text{ starts with a } 1 \text{ and ends with a different } 1\}.$$

Solution. The DFA rejects all strings that start with a zero, and then uses ϵ -transitions to check if the last symbol is a 1.

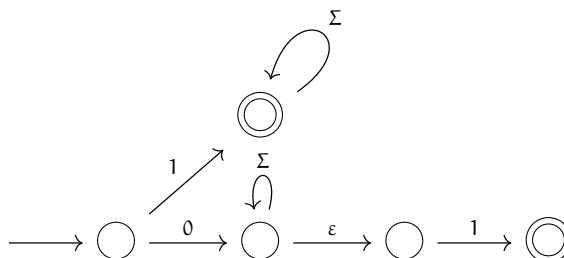


Problem 2. Construct an ϵ DFA to recognize the language

$$L_2 := \{w \in \{0, 1\}^* : w \text{ starts with } 1 \text{ or ends with } 1\}.$$

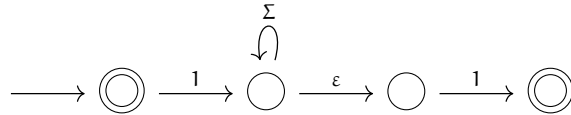
Hint: Sure you can do product construction. But can you make a smaller ϵ DFA?

Solution. The DFA accepts all strings that start with 1. If a string does not start with 1 it uses ϵ -transitions to check if the last symbol is a 1.



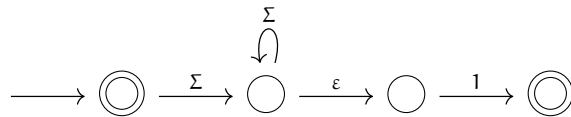
Problem 3. Construct an ϵ DFA to recognize the language that contains all binary strings that can be decomposed into substrings, where each substring starts with a 1 and ends with a different 1. In notation, L_3 is just the Kleene star of L_1 , that is, $L_3 := L_1^*$.

Solution. The DFA accepts the empty string, rejects all strings that start with a zero, and then uses ϵ -transitions to check if the last symbol is a 1.



Problem 4. Construct an ϵ DFA to recognize the language of the expression $((00)^*(0+1)^*1)^*$.

Solution. The DFA accepts the empty string, and rejects all strings that don't end with a 1.



Problem 5. *[To think about later]* Given an arbitrary ϵ DFA M , design another NFA $\hat{M}$ that recognizes the same language, but with a single accepting state.

Problem 6. *[To think about later]* For a string $w \in \{0,1\}^*$, $\text{rev}(w)$ is w written backwards (or ϵ if $w = \epsilon$). For a language L recognized by some DFA D , define $\text{rev}(L) := \{\text{rev}(x) : x \in L\}$. Design an ϵ DFA to recognize $\text{rev}(L)$.

Problem 5. *[Conceptual]* Recall the complement of an automatic language is still automatic. Can you repeat this argument for languages recognized by ϵ DFAs? Are languages recognized by ϵ DFAs closed under complement?